

Apple Inc. – Performance & Risk Analysis Report (1981–2025)

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1. Executive Summary

This report provides a concise performance and risk assessment of Apple Inc. over the period 1981-2025, using monthly return data merged with the Fama-French factor dataset. The objective is to evaluate Apple's long-term return profile, volatility behavior and sensitivity to key market factors.

Over the full sample, Apple delivered **exceptional long-term performance**, with a **CAGR of 18.55%**, significantly above broad-market benchmarks. However, this return profile came with **elevated volatility (43.39%)** and **deep drawdowns**, reflecting Apple's exposure to market cycles, product transitions and macroeconomic shocks.

Despite the high volatility, Apple's **Sharpe Ratio of 0.42** indicates a solid risk-adjusted return over the long run.

2. Performance Overview

2.1 Cumulative Return Growth

Apple's cumulative return trajectory shows **three distinct phases**:

- **1981-2004:** Low and inconsistent performance, reflecting early-stage business cycles and competitive pressure.
- **2004-2012:** Strong acceleration driven by the iPod, iPhone and ecosystem expansion.
- **2012-2025:** Sustained compounding with occasional corrections, supported by services revenue and global scale.

The long-term compounding effect is evident: despite early stagnation, Apple's cumulative return curve becomes exponential after 2005.

3. Risk Analysis

3.1 Rolling 12-Month Volatility

Volatility has been **structurally declining** over the last two decades. Key observations:

- **High volatility** during the early 1980s and dot-com era.
- **Sharp spikes** around 2000-2002 and 2008-2009.
- **Post-2010:** Volatility stabilizes at lower levels, reflecting Apple's transition to a mega-cap, diversified revenue model.

This trend indicates a shift from a high-uncertainty growth company to a more mature, stable cash-flow generator.

3.2 Drawdowns

Apple experienced **several deep drawdowns**, with the most severe exceeding **-79%**. Major drawdown periods align with:

- Early 2000s tech bubble
- 2008 financial crisis
- 2022-2023 macro tightening cycle

Despite these declines, Apple consistently recovered, demonstrating strong long-term resilience.

4. Factor Exposure Analysis

Using the merged Fama-French dataset, we evaluate Apple's relationship with key risk factors.

4.1 Market Excess Return (Mkt-RF)

The scatter plot shows a **clear positive relationship** between Apple's returns and market excess returns. This indicates:

- Apple behaves as a **high-beta growth stock**
- Market cycles significantly influence short-term performance

Apple's sensitivity to market risk is expected for a large technology company with global exposure.

4.2 Momentum Factor (Mom)

The relationship between momentum and Apple's returns is **weak but slightly positive**. Interpretation:

- Apple tends to benefit during momentum-driven markets
- But its performance is not purely momentum-dependent
- Fundamental catalysts (product cycles, earnings) play a larger role

4.3 Value Factor (HML)

The HML scatter plot shows **no meaningful correlation**. This is consistent with Apple's profile:

- Apple is **not a value stock**
- Its valuation is driven by growth, brand strength and ecosystem effects
- HML exposure is close to zero

5. Overall Assessment

Apple's long-term performance is characterized by:

- **Exceptional compounding returns**
- **High but declining volatility**

- **Strong sensitivity to market cycles**
- **Minimal exposure to value factors**
- **Moderate momentum characteristics**

From an investment perspective, Apple demonstrates the classic profile of a **high-quality growth company**: strong long-term returns, periodic deep drawdowns and performance driven by innovation cycles.

6. Conclusion

Apple's historical performance highlights the importance of long-term holding periods and the power of compounding in high-growth technology companies. While the stock has experienced significant volatility and drawdowns, its ability to recover and generate sustained returns underscores its resilience and strategic positioning.

This analysis demonstrates a structured approach to evaluating equity performance using return data, volatility metrics, and factor models, a methodology commonly applied in equity research and portfolio analysis.